

Parameter Estimation for Energy-Efficient 1-bit Bi-Static ISAC Network

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Abstract—The growing demand for robust, green, and wide-area sensing in applications like the low-altitude economy necessitates scalable, low-cost, and energy-efficient network architectures. To meet this need, this paper proposes a one-bit bi-static integrated sensing and communication (ISAC) system architecture designed to cover sensing blind spots in a cost-effective and energy-efficient manner. The system employs micro receiving nodes equipped with one-bit analog-to-digital converters to minimize hardware cost and power consumption. However, this design introduces the dual challenges: severe signal distortion from one-bit quantization and parameter ambiguity from clock asynchronism between the transmitter and receiver. These issues render conventional synchronization and parameter estimation methods ineffective. This paper presents a novel algorithm that directly tackles these challenges by processing the one-bit temporal data. Our method first estimates the direction of arrival (DOA) using the one-bit MUSIC algorithm, and leverages it to coherently accumulate signal energy in the spatial domain for each preamble. Aided by the DOA, a compressive sensing-based approach then estimates the combined delay and timing offset, followed by the extraction of the Doppler shift mixed with carrier frequency offset. By systematically processing these estimates, our algorithm obtains absolute target delays and relative Doppler shifts without requiring a dedicated reference path. The Cramér-Rao lower bound (CRLB) is derived, and simulations validate that the proposed method achieves high-accuracy parameter estimation, i.e., the optimization result reaches the CRLB. In addition, simulation under practical settings illustrates that the sensing range is extended from 400 m to 900 m, increased by 125%, demonstrating the feasibility of low-cost, one-bit bi-static ISAC systems.

Index Terms—Integrated sensing and communication, coarse quantization, passive sensing, clock asynchronism.

I. INTRODUCTION

AS THE advancement of vehicular technology rapidly expands from the ground to the air, the low-altitude economy, characterized by applications like urban air mobility, uncrewed logistics, and aerial surveillance, is emerging as a

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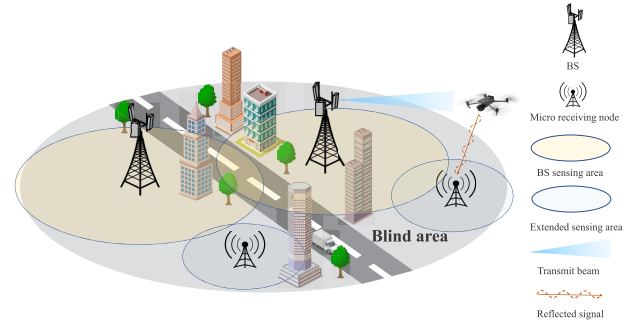


Fig. 1. Bi-static ISAC sensing scenario with one-bit reception nodes.

new frontier [1]. The proliferation of uncrewed aerial vehicles (UAVs) and other low-altitude aircraft in this space necessitates a robust, pervasive, and high-precision sensing network to ensure operational safety and efficient traffic management. However, deploying dedicated radar systems on a massive scale for comprehensive airspace coverage is prohibitively expensive and leads to an increasingly congested radio frequency (RF) spectrum.

To address these challenges, integrated sensing and communication (ISAC) has emerged as a promising technology that integrates sensing and communication functionalities into a single system using shared hardware and signal resources [2], [3], [4]. This dual-functional approach not only alleviates spectrum scarcity [5], [6] but also reduces the overall cost and power consumption of the system [7], [8], making it a key enabler for future intelligent transportation systems. While a network of ISAC-enabled base stations (BSs) can provide wide-area coverage, sensing blind spots inevitably exist, particularly at the cell edges or in areas occluded by complex urban environments, as illustrated in Fig. 1.

To overcome the coverage limitations of mono-static systems, distributed architectures such as the Perceptive Mobile Network (PMN) have been proposed [41], [42]. The PMN framework integrates radar sensing into cellular networks, enabling bi-static and multi-static sensing by exploiting cooperative processing. However, such architectural studies typically rely on the assumption of perfect synchronization between distributed nodes and utilize high-precision hardware. Implementing these architectures in cost-sensitive scenarios with massive node deployment remains a challenge. Specifically, while deploying standard receivers with high-precision

ADCs could theoretically cover blind spots, achieving the high angular resolution required to distinguish closely spaced targets necessitates scaling up the antenna array [9]. With high-precision ADCs, this results in prohibitive power consumption and hardware costs, hindering massive deployment.

To address these problems, a bi-static ISAC system architecture specifically designed to cover sensing blind spots is proposed in this paper. In the proposed system, passive reception nodes are strategically placed in the blind-spot areas, forming a bi-static configuration with the active ISAC BSs. Since these passive nodes are often closer to the targets than the backscattering path to the active BS, they can achieve a sufficient signal-to-noise ratio (SNR) for reliable detection. Capitalizing on this favorable link budget, we further minimize the hardware cost and complexity by equipping these reception nodes with one-bit analog-to-digital converters (ADCs). As an extreme case of coarse quantization, one-bit quantization represents the analog signal with only a single bit [10], [11], [12], [13], [14], essentially simplifying the hardware to that of a simple comparator. This design choice trades amplitude resolution for spatial resolution: it enables the use of larger antenna arrays for high angular precision while employing one-bit quantization to drastically reduce hardware complexity, e.g., eliminating Automatic Gain Control (AGC), and energy consumption [15], [16]. This makes the proposed architecture exceptionally cost-friendly and power-efficient, paving the way for ubiquitous high-resolution sensing.

Despite these benefits, this cost-effective design introduces two profound challenges for parameter estimation. The first is the severe non-linearity from one-bit quantization, which significantly degrades the signal quality. To mitigate such degradation, extensive research has addressed signal processing for one-bit systems. In radar detection, theoretical analyses have quantified the performance loss of one-bit quantization [36], while techniques like time-varying thresholds have been proposed to recover amplitude information [37]. Furthermore, sparse recovery frameworks have been widely adopted to enhance estimation precision for PMCW radars [24] and DoA estimation [26]. In the communication domain, deep learning approaches have also been applied for robust one-bit detection [27]. Building on these foundations, recent works have begun to address specific hardware constraints in ISAC architectures, including low-complexity MIMO-DFRC systems via spatial modulation [38], waveform design under limited feedforward channels [39], and bit-constrained receivers using task-based quantization [40].

The second, inherent to the bi-static sensing scenario, is clock asynchronism [17]. In conventional communication systems, the timing offset (TO) can be absorbed into the channel estimate and the carrier frequency offset (CFO) can be compensated, effectively eliminating their impact. In stark contrast, for sensing, the primary goal is to measure the channel itself. Therefore, these same offsets introduce a critical measurement ambiguity between the clock error and the desired parameters (e.g., target delay and Doppler), which degrades estimation accuracy. This inherent ambiguity renders standard tracking methods unsuitable. Consequently, our algorithm must confront the raw, uncompensated error at each preamble, whose

apparent characteristics are fast-varying due to unsuppressed clock jitter and inter-preamble drift [21].

Several techniques have been developed to solve the problem, by exploiting the characteristics of the received signal from different antennas [18], [19], the cross-antenna cross-correlation (CACC) [18] and cross-antenna signal ratio (CASR) [19] algorithms are proposed, which remove the unsynchronized term via correlating or dividing the signal between different antennas. In addition, some cancellation methods are proposed to estimate and compensate the unsynchronized term, which could be applied in a single-antenna system [20]. Though these methods perform well for high-resolution quantization, they cannot handle the coarse quantization case, as the time-domain one-bit quantization deteriorates the temporal signal severely, and the frequency-domain channel state information could be hard to obtain with a simple fast Fourier transform (FFT).

In this paper, in order to cover the sensing blind area cost-effectively and strengthen the sensing ability of the network, a low-complexity bi-static ISAC system is proposed, where one-bit reception nodes are used for range extension. In addition, a clock-asynchronism-aware one-bit parameter estimation method is proposed for this bi-static ISAC system. This method discovers the characteristic of the clock asynchronism, i.e., offsets between antennas are the same, integrating energy from the spatial domain, and decouples the TO-caused phase shift in a two-stage manner to extract the Doppler component. Specifically, the proposed method estimates the absolute value of the direction of arrival (DOA) first via one-bit multiple signal classification (MUSIC). Based on this, a DOA-aware compressive sensing method is introduced for delay plus TO (DPT) estimation, which also accumulates the energy in the spatial domain. After that, the Doppler frequency is decoupled with delay, and the CFO mixed Doppler component could be extracted via iterative gradient descent. This method reaches the Cramér-Rao lower bound (CRLB) with a relatively low-complexity algorithm. Simulations demonstrate that the proposed method could enlarge the sensing range dramatically, from 400 m to 900 m, improved by 125%, and extend the sensing area from 280,000 m² to 640,000 m² with the aid of a low-complexity, low-cost one-bit micro receiving node, which could cover the blind spot of the ISAC system effectively.

The main contributions of the paper are summarized as follows:

- 1) *A low-complexity bi-static system is proposed:* A one-bit ADC-based low-cost bi-static system is proposed, which reduces the complexity and cost of the receiver. Based on this architecture, the ISAC network could extend the sensing range, cover the blind area, and strengthen the sensing ability.
- 2) *A clock-asynchronism-aware one-bit estimation algorithm is proposed:* A parameter estimation algorithm for bi-static asynchronous systems is proposed. This algorithm decouples the high-dimensional parameters in a low-complexity manner and achieves the CRLB for optimization variables, with only a minor performance gap, i.e., 1.96 dB, between single-bit and high-precision quantization in theory.

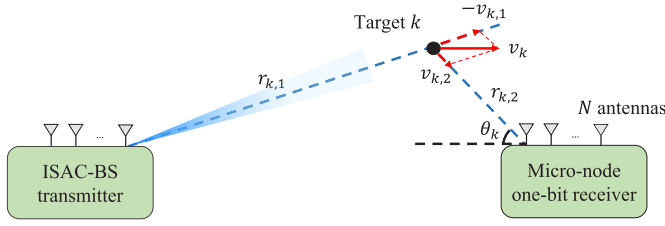


Fig. 2. Delay and doppler frequency shift for passive sensing.

- 3) *Simulations are provided to demonstrate the effectiveness of the algorithm:* Parameter estimation results are provided to demonstrate the effectiveness of the algorithm. It is shown that the probability of successful sensing improves from 31.2% to 63.8%. And the sensing range extends from 400 m to 900 m, increased by 125% with the aid of a low-complexity, low-cost one-bit micro receiving node under practical BS settings.

The rest of the paper is organized as follows: Section II introduces the signal and system model under clock asynchronism, and an optimization problem is formulated to extract the useful information from one-bit measurements. Section III presents the main algorithm and analyzes its computational complexity. Section IV discusses the theoretical performance and derives the CRLB. Simulations and numerical experiments are presented in Section V. Finally, Section VI concludes the whole paper.

Notations: Bold-face lower case and upper case letters denote vectors and matrices, respectively. $\mathbb{C}$ and $\mathbb{R}$ denote the complex-value and real-value. $\mathbb{C}^{P \times N}$ and $\mathbb{C}^{N \times 1}$ denote the spaces of $P \times N$ and $N \times 1$ complex matrices and vectors, respectively. $j = \sqrt{-1}$ denotes the imaginary unit. The operations $(\cdot)^T$, $(\cdot)^H$, $\otimes$, $\odot$, $*$, and $|\cdot|$ represent transpose, conjugate transpose, Kronecker product, Hadamard product, convolution, and absolute value, respectively. Real and imaginary parts are denoted as $\Re\{\cdot\}$ and $\Im\{\cdot\}$, respectively. $\mathbb{P}$ and $\mathbb{E}$ denote the mathematical probability and expectation. $\|\cdot\|_2$ denotes the ℓ_2 -norm. $\mathcal{O}(\cdot)$ denotes the standard big-O notation.

II. SYSTEM MODEL AND PROBLEM FORMULATION

This section introduces the main scenario and the received signal model for a multi-antenna array with clock asynchronism and coarse quantization, i.e., one-bit quantization.

A. Scenario

Consider a passive-sensing scenario in Fig. 2, where a BS is sweeping the transmit beam for target sensing. Micro-reception nodes specified for signal receiving are installed to extend the sensing range of the 5G cellular network. Assume there are K targets in the sweeping area. Without loss of generality, we consider passive sensing of the k -th target. The distances from the k -th target to the transmitter and receiver are $r_{k,1}$ and $r_{k,2}$, respectively. And the corresponding radial velocities in the direction of BSs are $v_{k,1}$ and $v_{k,2}$. The micro-node receiver is equipped with N antennas which are uniformly spaced with the interval of $\lambda/2$, i.e., a uniform

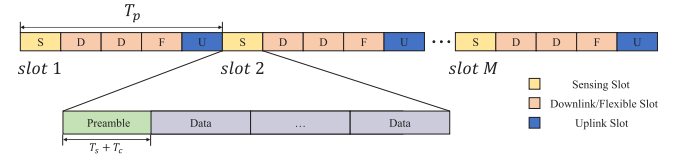


Fig. 3. The proposed 5G-A TDD frame structure.

linear array, where λ is the wavelength of the signal. The incident angle of the echo signal is θ_k , and the signal is quantized with one-bit ADCs to save the additional hardware and energy costs. However, the severe non-linearity of the one-bit quantization makes parameter estimation challenging. In addition, clock asynchronism between BSs results in time-varying TO and CFO, which leads to ambiguity in target sensing [21]. In this case, the existence of these imperfect components further brings challenges for one-bit sensing in the passive-mode ISAC system. It is crucial to discover a method for accurate target sensing in this scenario.

B. Received Signal Without Clock Asynchronism

Based on the scenario model, the signal model is constructed first. We consider a 5G-Advanced (5G-A) time division duplex (TDD) frame structure, as illustrated in Fig. 3. The frame consists of distinct slots: Sensing (S), Downlink (D), Flexible (F), and Uplink (U). The proposed micro-nodes perform passive sensing primarily during the dedicated sensing slots.¹ The ISAC signal is OFDM-based, and preambles with the pulse repetition interval (PRI) of T_p are used for sensing. It is also assumed that there are M preambles within a coherent processing interval. Then the m -th preamble signal is expressed as

$$s_m(t) = \sum_{p=0}^{P-1} x_{p,m} e^{j2\pi(f_c + p\Delta f)t} \text{rect}\left(\frac{t - mT_p}{T_s + T_c}\right),$$

$$m = 0, 1, \dots, M-1, \quad (1)$$

where P subcarriers are considered with the interval of Δf , $x_{p,m}$ is the pre-known pilot on the p -th subcarrier of the m -th preamble, f_c is the carrier frequency, $T_s = 1/\Delta f$ is the duration of the OFDM symbol, and $\text{rect}(\cdot)$ is the rectangle window function, i.e. $\text{rect}\left(\frac{t}{T_s + T_c}\right) = 1$ when $t \in [0, T_s + T_c]$, corresponding to the period of an OFDM symbol with cyclic prefix (CP).

Furthermore, based on the antenna model, the steering vector can be represented as

$$\mathbf{b}(\theta_k) = [1, e^{-j\pi \sin(\theta_k)}, \dots, e^{-j\pi(N-1) \sin(\theta_k)}]^T \in \mathbb{C}^{N \times 1}. \quad (2)$$

¹Although this work focuses on sensing for blind-spot coverage, the proposed one-bit micro-nodes are fully capable of serving as communication receivers. During communication slots, information can be recovered using established one-bit signal processing techniques, such as Bussgang-based LMMSE estimators [10], Near-Maximum Likelihood detectors [11], or EM-based algorithms [27]. Since these solutions are well-documented, this paper concentrates exclusively on the unsolved challenge of sensing parameter estimation under asynchronous clock constraints.

Then, the multi-path channel in the time domain is denoted as

$$\tilde{\mathbf{h}}(t) = \sum_{k=0}^{K-1} \tilde{\alpha}_k G_{t,k} \delta(t - \tau_k) \mathbf{b}(\theta_k), \quad (3)$$

where $\tilde{\alpha}_k$, $G_{t,k}$, and τ_k are the complex channel path loss, antenna gain from the transmit beampattern, and signal propagation delay for the k -th target, respectively. By doing a convolution with the transmit signal, the noise-free RF signal can be written as

$$\begin{aligned} \mathbf{y}_m(t) &= s_m(t) * \tilde{\mathbf{h}}(t) \\ &= \sum_{k=0}^{K-1} \sum_{p=0}^{P-1} \tilde{\alpha}_k G_{t,k} x_{p,m} e^{j2\pi(f_c + p\Delta f)(t - \tau_k)} \mathbf{b}(\theta_k) \\ &= \sum_{k=0}^{K-1} \sum_{p=0}^{P-1} \alpha_k x_{p,m} e^{j2\pi(f_c + f_{D,k})t} e^{j2\pi p\Delta f(t - \tau_k)} \mathbf{b}(\theta_k), \end{aligned} \quad (4)$$

where the rectangle window function is omitted for ease of expression, because the delay is assumed to be in the duration of CP, the desired mixed signal is inside the rectangle window. Furthermore, given the light speed c , the delay and Doppler frequency could be expressed as $\tau_k = r_k/c$ and $f_{D,k} = v_k f_c/c$, where $r_k = r_{k,1} + r_{k,2}$ and $v_k = v_{k,1} + v_{k,2}$ are the aggregate distance and radial relative velocity of the non-line-of-sight (NLOS) path reflected from the k -th target, respectively. And $\alpha_k = \tilde{\alpha}_k G_{t,k} e^{-j2\pi f_c r_k/c}$ is the equivalent path loss.

The continuous signal is then down-converted via the reception node's LO and discretized at ADCs with a Nyquist sampling rate. After that we could formulate a noise-free baseband matrix $\mathbf{Y}_m \in \mathbb{C}^{P \times N}$, which is expressed as

$$\mathbf{Y}_m = \sum_{k=0}^{K-1} \alpha_k v_m(f_{D,k}) \mathbf{F}^H(\mathbf{x}_m \odot \mathbf{r}(\tau_k)) \mathbf{b}(\theta_k)^T, \quad (5)$$

where $\mathbf{F}^H \in \mathbb{C}^{P \times P}$ is the unitary discrete inverse Fourier transform matrix with the entries $[\mathbf{F}]_{a,b} = e^{j2\pi(a-1)(b-1)/P} / \sqrt{P}$ for $a, b = 0, \dots, P-1$; $\mathbf{x}_m$ is the circular complex Gaussian distributed pilot vector following $\mathbf{x}_m \sim \mathcal{CN}(\mathbf{0}, \sigma_x^2 \mathbf{I}_P)$. The delay and Doppler encoded vectors are denoted as

$$\mathbf{r}(\tau_k) = [1, e^{-j2\pi\Delta f\tau_k}, \dots, e^{-j2\pi(P-1)\Delta f\tau_k}]^T, \quad (6)$$

and

$$\mathbf{v}(f_{D,k}) = [1, e^{j2\pi f_{D,k}T_p}, \dots, e^{j2\pi f_{D,k}(M-1)T_p}]^T, \quad (7)$$

respectively, where $v_m(f_{D,k})$ is the m -th element of $\mathbf{v}(f_{D,k})$.

C. Received Signal With Clock Asynchronism

Notice that while the signal model in (5) geometrically resembles a mono-static model, the underlying hardware architecture introduces a fundamental challenge. In mono-static radar, the transmitter and receiver share a common LO, ensuring inherent synchronization. In contrast, the proposed bi-static architecture employs spatially separated nodes with independent, free-running LOs. This hardware separation inevitably introduces time-varying TO and CFO, as modeled in this subsection [22], [23]. These asynchronous terms couple with the target parameters, creating a non-linear parameter ambiguity problem that renders conventional mono-static estimation algorithms ineffective.

1) *TO Mixed Steering Vector*: Because of the clock dithering, the TO-free delay-related steering vector $\mathbf{r}(\tau_k)$ is mixed with a time-varying component. The m -th TO steering vector is expressed as

$$\delta_{\tau,m} = [1, e^{-j2\pi\Delta f\delta_{\tau,m}}, \dots, e^{-j2\pi(P-1)\Delta f\delta_{\tau,m}}]^T, \quad (8)$$

where $\delta_{\tau,m}$ changes with preambles. In the case, the delay encoded vector is coupled with preambles and the m -th DPT component is defined as $\tilde{\tau}_{k,m} = \tau_k + \delta_{\tau,m}$, whose steering vector $\mathbf{r}(\tilde{\tau}_{k,m}) \in \mathbb{C}^{P \times 1}$ is given by

$$\mathbf{r}(\tilde{\tau}_{k,m}) \triangleq \mathbf{r}(\tau_k) \odot \delta_{\tau,m}. \quad (9)$$

2) *CFO Mixed Steering Vector*: Because of the carrier frequency offset of two LOs, the Doppler frequency steering vector $\mathbf{v}(f_{D,k})$ is also mixed with a time-varying CFO component, which is denoted as

$$\delta_{f_D} = [1, \dots, e^{j2\pi m\delta_{f,m}T_p}, \dots, e^{j2\pi(M-1)\delta_{f,m-1}T_p}]^T, \quad (10)$$

where $\delta_{f,m}$ also changes with preambles. Then, the m -th CFO mixed Doppler component is written as $\tilde{v}_m(f_{D,k}) = e^{j2\pi(f_{D,k} + \delta_{f,m})mT_p}$, which is the m -th element of the CFO mixed Doppler frequency steering vector $\tilde{\mathbf{v}}(f_{D,k}) \in \mathbb{C}^{M \times 1}$ is given by

$$\tilde{\mathbf{v}}(f_{D,k}) \triangleq \mathbf{v}(f_{D,k}) \odot \delta_{f_D}. \quad (11)$$

Then, the received signal with clock asynchronism, denoted by $\tilde{\mathbf{Y}}_m$, is expressed as

$$\tilde{\mathbf{Y}}_m = \sum_{k=0}^{K-1} \alpha_k \tilde{v}_m(f_{D,k}) \mathbf{F}^H(\mathbf{x}_m \odot \mathbf{r}(\tilde{\tau}_{k,m})) \mathbf{b}(\theta_k)^T. \quad (12)$$

After adding the noise in the receiver and quantized by the one-bit ADCs, the discrete signal $\mathbf{Z}_m \in \mathbb{C}^{P \times N}$ is given by

$$\mathbf{Z}_m = \mathcal{Q}(\tilde{\mathbf{Y}}_m + \mathbf{W}_m), \quad (13)$$

where the elements of $\mathbf{W}_m$ are independent circular complex Gaussian noise with zero mean and variance σ_w^2 , and $\mathcal{Q}(\cdot)$ is the one-bit quantization operator, represented as

$$\mathcal{Q}(x) = \text{sign}(\Re\{x\}) + j \cdot \text{sign}(\Im\{x\}), \quad (14)$$

with $\text{sign}(\cdot)$ as a comparing function defined as

$$\text{sign}(x) = \begin{cases} 1 & x \geq 0, \\ -1 & x < 0. \end{cases} \quad (15)$$

Based on the coarsely quantized asynchronous signal model, the problem is formulated in the next section.

D. Problem Formulation

This section formulates the parameter estimation problem of one-bit measurements, the unknown parameters are denoted as $\boldsymbol{\beta} = [\tau^T, f_D^T, \theta^T, \alpha_R^T, \alpha_I^T, \delta_\tau^T, \delta_f^T]^T \in \mathbb{R}^{(5K+2M) \times 1}$, where

$$\boldsymbol{\tau} = [\tau_0, \dots, \tau_{K-1}]^T, \quad (16a)$$

$$\mathbf{f}_D = [f_{D,0}, \dots, f_{D,K-1}]^T, \quad (16b)$$

$$\boldsymbol{\theta} = [\theta_0, \dots, \theta_{K-1}]^T, \quad (16c)$$

$$\alpha_R = [\Re\{\alpha_0\}, \dots, \Re\{\alpha_{K-1}\}]^T, \quad (16d)$$

$$\alpha_I = [\Im\{\alpha_0\}, \dots, \Im\{\alpha_{K-1}\}]^T, \quad (16e)$$

$$\delta_\tau = [\delta_{\tau,0}, \dots, \delta_{\tau,M-1}]^T, \quad (16f)$$

$$\delta_f = [\delta_{f,0}, \dots, \delta_{f,M-1}]^T. \quad (16g)$$

Note that only τ , f_D , and θ are parameters of interest; this paper aims to estimate these unknown variables from the one-bit measurements.

To obtain a high-precision estimation of parameters, maximum-likelihood (ML) estimation is applied, which is widely used in one-bit estimation [24]. Further reshape the one-bit matrix $\mathbf{Z}_m \in \mathbb{C}^{P \times N}$ into a vector $\mathbf{z}_m \in \mathbb{C}^{PN \times 1}$ and stack all M preambles into $\mathbf{z} = [\mathbf{z}_0^T, \dots, \mathbf{z}_{M-1}^T]^T \in \mathbb{C}^{PNM \times 1}$, define the elements of $\mathbf{z}$ as z_i , because the independent noise is Gaussian distributed, the probability density function of the pre-quantized term $r_i = \tilde{y}_i + w_i$ is

$$\mathbb{P}(r_i|\tilde{y}_i) = \frac{1}{\pi\sigma_w^2} \exp\left(-\frac{\|r_i - \tilde{y}_i\|_2^2}{\sigma_w^2}\right). \quad (17)$$

After quantization, if $\Re\{z_i\} = 1$, the probability is calculated as

$$\begin{aligned} \mathbb{P}(\Re\{z_i\} = 1|\tilde{y}_i) &= \int_0^\infty \frac{1}{\sqrt{\pi\sigma_w^2}} \exp\left(-\frac{(\Re\{r_i\} - \Re\{\tilde{y}_i\})^2}{\sigma_w^2}\right) dr_i \\ &= \int_{-\Re\{\tilde{y}_i\}}^\infty \frac{1}{\sqrt{\pi\sigma_w^2}} \exp\left(-\frac{t^2}{\sigma_w^2}\right) dt \\ &\triangleq \Phi_w(\Re\{\tilde{y}_i\}), \end{aligned} \quad (18)$$

where $\Phi_w(x) = \int_{-\infty}^x \frac{1}{\sqrt{\pi\sigma_w^2}} \exp\left(-\frac{t^2}{\sigma_w^2}\right) dt$ is the cumulative Gaussian distribution. The result for $\Re\{z_i\} = -1$ case is

$$\mathbb{P}(\Re\{z_i\} = -1|\tilde{y}_i) = 1 - \Phi_w(\Re\{\tilde{y}_i\}) = \Phi_w(-\Re\{\tilde{y}_i\}). \quad (19)$$

Therefore, the probability density function (PDF) for quantized data is written as

$$\mathbb{P}(z_i|\tilde{y}_i) = \Phi_w(\Re\{z_i\}\Re\{\tilde{y}_i\})\Phi_w(\Im\{z_i\}\Im\{\tilde{y}_i\}), \quad (20)$$

and the vector form is expressed as

$$\mathbb{P}(\mathbf{z}|\tilde{\mathbf{y}}) = \prod_{i=1}^{PNM} \mathbb{P}(z_i|\tilde{y}_i). \quad (21)$$

The goal is to maximize the likelihood function, which is represented as

$$\begin{aligned} \ln \mathbb{P}(\mathbf{z}|\tilde{\mathbf{y}}) &= \sum_{i=1}^{PNM} \ln \mathbb{P}(z_i|\tilde{y}_i) \\ &= \sum_{i=1}^{PNM} \left(\ln \Phi_w(\Re\{z_i\}\Re\{\tilde{y}_i\}) + \ln \Phi_w(\Im\{z_i\}\Im\{\tilde{y}_i\}) \right), \end{aligned} \quad (22)$$

where $\tilde{\mathbf{y}}$ is determined by the unknown variables $\boldsymbol{\beta}$, comprising of interested and uninterested values (12), i.e.

$$\arg \max_{\tau, f_D, \theta} \left\{ \max_{\alpha_R, \alpha_I, \delta_\tau, \delta_f} \ln \mathbb{P}(\mathbf{z}|\boldsymbol{\beta}) \right\}. \quad (23)$$

This optimization problem presents a fundamental challenge distinct from standard synchronous radar. In conventional settings, delay and Doppler are deterministic parameters that

can be estimated separately in the frequency and time domains. However, in our bi-static architecture with independent clocks, the TO and CFO vary across preambles. This introduces a severe parameter coupling problem: the observed delay and phase fluctuations are an inseparable mixture of target motion and random clock jitter. Consequently, the target parameters $\{\tau, f_D, \theta\}$ are entangled with the high-dimensional nuisance parameters $\{\delta_{\tau,m}, \delta_{f,m}\}_{m=0}^{M-1}$.

Specifically, the observed delay estimates $\tilde{\tau}_{k,m} = \tau_k + \delta_{\tau,m}$ are mixed with TO components. Consequently, the result of the likelihood function is unchanged with varying τ_k and $\delta_{\tau,m}$, as far as keeping the sum value $\tilde{\tau}_{k,m}$ fixed, making range estimation ambiguous. Doppler frequency estimation faces the same condition. Given the mixed term $\alpha_k \tilde{v}_m(f_{D,k}) = \alpha_k e^{j2\pi(f_{D,k} + \delta_{f,m})mT_p}$, the CFO component $e^{j2\pi\delta_{f,m}mT_p}$ is a random phase shift. The likelihood function could be maximized via optimizing the composite term $\alpha_k \tilde{v}_m(f_{D,k})$, while the absolute values α_k and $f_{D,k}$ are undetermined with different $\delta_{f,m}$. This inherent ambiguity makes a direct joint search over this $(3 + 2M)$ -dimensional space computationally prohibitive and prone to local minima due to the non-convex landscape of one-bit quantization.

To cope with the above problems, the TO and CFO mixed terms are estimated first by solving the following optimization problem

$$\arg \max_{\tilde{\boldsymbol{\tau}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\nu}}} \ln \mathbb{P}(\mathbf{z}|\boldsymbol{\beta}), \quad (24)$$

where $\tilde{\boldsymbol{\tau}} \in \mathbb{R}^{KM \times 1}$ and $\hat{\boldsymbol{\nu}} \in \mathbb{C}^{KM \times 1}$ are two vectors consisting KM variables, represented as

$$\tilde{\boldsymbol{\tau}} = [\tilde{\tau}_{0,0}, \dots, \tilde{\tau}_{K-1,M-1}]^T, \quad (25)$$

$$\hat{\boldsymbol{\nu}} = [\alpha_0 \tilde{v}_0(f_{D,0}), \dots, \alpha_{K-1} \tilde{v}_{M-1}(f_{D,K-1})]^T, \quad (26)$$

respectively. Notice the dimensions of the unknown variables change from $5K + 2M$ to $(2M + 1)K$. Because the TO and CFO for each preamble m are independent, it could be regarded as if there are $2K$ unrelated variables for each preamble m , i.e., K delay encoded components and K complex amplitudes. Until now, the optimization problem (24) could be simplified into M sub-optimization problem

$$\arg \max_{\tilde{\tau}_m, \hat{\nu}_m} \left\{ \max_{\theta} \ln \mathbb{P}(\mathbf{z}_m|\boldsymbol{\beta}) \right\}. \quad (27)$$

This optimization problem (27) could be solved by iteratively estimating θ with PNM one-bit measurements and estimating $\tilde{\tau}_m, \hat{\nu}_m$ with PN one-bit measurements for M preambles via sparse recovery algorithm. However, in this paper, θ is estimated via one-bit MUSIC to ease the calculation pressure and lower the complexity of the algorithm. After optimization, the absolute value of delay τ is obtained via calculating the average value of M preambles τ_m and the absolute value of Doppler frequency f_D could be calculated via correlating with a static reference path, i.e. static targets or clutters. The detail parameter estimation algorithm is provided in the sequel.

III. ONE-BIT SENSING WITH CLOCK ASYNCHRONISM

In this section, we present our clock-asynchronism-aware parameter estimation algorithm, based on a strategic, multi-stage decoupling framework. Since the clock asynchronism

introduces a common phase offset across all antennas, the spatial domain is unaffected. Therefore, the DOA is estimated first via the low-complexity one-bit MUSIC algorithm. Then, by coherently accumulating this spatial gain, we process each preamble individually using a sparse recovery approach. This is necessary because the TO and CFO are coupled; we must first estimate the combined DPT before the Doppler-plus-CFO component can be extracted. Finally, with the effects of delay and TO removed, the CFO-mixed Doppler component is extracted from the reconstructed signal, and based on which, by doing correlation between these components, the relative Doppler shifts of targets can be obtained.

A. DOA Estimation via One-Bit MUSIC

DOA estimation is made first because it is naturally decoupled from range and Doppler components, which are coupled due to clock asynchronism. It is verified in [25], [43] that the MUSIC technique could be directly applied in one-bit DOA estimation, termed as one-bit MUSIC, which is widely used in DOA estimation.

It is worth noting that while a direct Maximum Likelihood (ML) estimation would be theoretically optimal, it involves a computationally prohibitive joint search over a high-dimensional space involving time-varying nuisance parameters. To address this, we employ one-bit MUSIC for dimensionality reduction. Although subspace-based estimators may exhibit bias compared to ML estimators, especially under coarse quantization, they offer a significant advantage in computational efficiency. Furthermore, the primary role of the DOA estimate in our framework is to facilitate coherent spatial accumulation. Since the array gain is relatively insensitive to small angular deviations, the estimation accuracy of one-bit MUSIC is sufficient to ensure robust SNR boosting for the subsequent stages.

Stacking all of M preambles' $P \times N$ one-bit data yields

$$\mathbf{Z} = [\mathbf{Z}_0^T, \mathbf{Z}_1^T, \dots, \mathbf{Z}_{M-1}^T]^T \in \mathbb{C}^{PM \times N}. \quad (28)$$

Consequently, the correlation matrix could be calculated as

$$\mathbf{R}_z = \frac{1}{PM} \mathbf{Z}^H \mathbf{Z}, \quad (29)$$

which is equivalent to sampling PM points in the temporal domain with N antennas, as the DOA estimation is decoupled with range and Doppler components. In this case, via performing eigenvalue decomposition to $\mathbf{R}_z$, which is expressed as

$$[\mathbf{U}, \mathbf{\Sigma}] = \text{eig}(\mathbf{R}_z). \quad (30)$$

The steering vector of DOA could be obtained in the orthogonal eigen matrix $\mathbf{U}$ comprising columns of eigenvectors. Select the eigenvectors associated with the smallest $N-K$ eigenvalues as the noise subspace $\mathbf{U}_{K+1:N}$ and formulate the spectrum as

$$G(\theta) = \frac{1}{\mathbf{b}^H(\theta) \mathbf{U}_{K+1:N} \mathbf{U}_{K+1:N}^H \mathbf{b}(\theta)}. \quad (31)$$

With which the DOA could be estimated by searching the spectrum with different θ and finding the result corresponding to the K maximum peaks, i.e.

$$\{\hat{\theta}_k\}_{k=0}^{K-1} = \arg \max_{\theta} |G(\theta)|. \quad (32)$$

B. Doppler Decoupled Range Estimation

In this subsection, a sparse recovery problem is formulated to estimate the DPT term $\tilde{\tau}_{k,m}$ for each preamble using the steepest descent method. Because of the clock asynchronism, the delay and Doppler frequency terms are mixed with dithering components, i.e., TO and CFO. These time-varying components couple the delay and Doppler frequency terms within different preambles m , i.e., delay and Doppler frequency are varying with m simultaneously, as shown in (12). To decouple these components, the DPT term $\tilde{\tau}_{k,m}$ is first estimated for each preamble m with the help of the estimated DOA, which could be used for coherent accumulation in the spatial domain. For the ease of calculation, it is assumed that the transmitted symbols for each preamble are fixed, i.e. $\mathbf{x}_m = \mathbf{x}$.

Based on the likelihood function (27), the goal is to find the appropriate DPT $\tilde{\tau}_m$ to maximize the equation for the m -th preamble with estimated $\hat{\theta}$ and unknown variable $\alpha \tilde{\mathbf{v}}_m(f_D)$, i.e.

$$\max_{\tilde{\tau}_m} \ln \mathbb{P}(\mathbf{z}_m | \tilde{\mathbf{y}}_m), \quad (33)$$

where $\tilde{\mathbf{y}}_m \in \mathbb{C}^{PN \times 1}$ is written as

$$\tilde{\mathbf{y}}_m = \sum_{k=0}^{K-1} \alpha_k \tilde{\mathbf{v}}_m(f_{D,k}) \mathbf{b}(\theta_k) \otimes (\mathbf{F}^H(\mathbf{x} \odot \mathbf{r}(\tilde{\tau}_{k,m}))). \quad (34)$$

To this end, this sub-optimization problem could be rewritten as

$$\max_{\tilde{\tau}_m} \sum_{i=1}^{PN} \left(\ln \Phi_w(\Re\{z_{m,i}\} \Re\{\tilde{y}_{m,i}\}) + \ln \Phi_w(\Im\{z_{m,i}\} \Im\{\tilde{y}_{m,i}\}) \right), \quad (35)$$

where $\tilde{y}_{m,i}$ is the i -th element of $\tilde{\mathbf{y}}_m$. Although the function $-\ln \Phi_w(x)$ is convex, $\tilde{y}_{m,i}$ is not monotonically increasing or decreasing with respect to $\tilde{\tau}_{k,m}$, while the maximum value could be found by searching all of the possible values for $\tilde{\tau}_{k,m}$. In this case, the optimization problem could be further rewritten as a sparse recovery problem with the dictionary matrix denoted as

$$\mathbf{A} = \mathbf{B} \otimes \mathbf{R} \in \mathbb{C}^{PN \times KG}, \quad (36)$$

where $\mathbf{B} = [\mathbf{b}(\hat{\theta}_0), \dots, \mathbf{b}(\hat{\theta}_{K-1})] \in \mathbb{C}^{N \times K}$ is the reconstructed steering vectors based on the estimated DOA, which is used for coherent accumulation,

$$\mathbf{R} = [\mathbf{F}^H(\mathbf{x} \odot \mathbf{r}(\tau_0)), \dots, \mathbf{F}^H(\mathbf{x} \odot \mathbf{r}(\tau_{G-1}))] \in \mathbb{C}^{P \times G} \quad (37)$$

is the estimation matrix with G grids corresponding to different candidate delay values on the grid, then (34) could be transformed to

$$\tilde{\mathbf{y}}_m = \mathbf{A} \mathbf{u}_m, \quad (38)$$

where $\mathbf{u}_m \in \mathbb{C}^{KG \times 1}$ is the sparse vector, with the corresponding non-zero elements equal to $\alpha_k \tilde{\mathbf{v}}_m(f_{D,k})$. Then, (33) could be transformed and formulated as a sparse recovery problem

$$\min_{\mathbf{u}_m} -\ln \mathbb{P}(\mathbf{z}_m | \mathbf{A} \mathbf{u}_m) + \lambda \|\mathbf{u}_m\|_1 \quad (39)$$

which is convex [26], [27] and could be solved via CVX toolbox or steepest gradient descent method with gradient

$$\Delta \mathbf{u}_m = -\frac{1}{2} \mathbf{A}^H \{f'(\Re\{\mathbf{z}_m\}) \odot \Re\{\mathbf{A}\mathbf{u}_m\}\} \odot \Re\{\mathbf{z}_m\} - \frac{j}{2} \mathbf{A}^H \{f'(\Im\{\mathbf{z}_m\}) \odot \Im\{\mathbf{A}\mathbf{u}_m\}\} \odot \Im\{\mathbf{z}_m\} + \frac{\lambda}{2} \mathbf{D}\mathbf{u}_m, \quad (40)$$

where $\mathbf{D} = \text{diag}\left\{\frac{1}{|u_{m,i}|}\right\}_{i=1}^{PN}$ and $f'(x) = \phi_w(x)/\Phi_w(x)$, where $\phi_w(x) = \frac{1}{\sqrt{\pi\sigma_w^2}} \exp\left(-\frac{x^2}{\sigma_w^2}\right)$ is the derivative of $\Phi_w(x)$. And $\{\hat{\tau}_{k,m}\}_{k=0}^{K-1}$ is obtained from the non-zero index of the sparse vector $\mathbf{u}_m$. The gradient is derived by minimizing the penalized negative log-likelihood function. The detailed derivation involves the derivative of the probit likelihood and the handling of the regularization term, which is provided in Appendix A. It is also important to note that this grid-based approach assumes a sufficiently dense grid to mitigate the off-grid problem, where true delays may not fall exactly on the grid points. The choice of grid resolution represents a trade-off between mitigating this basis mismatch effect and maintaining computational feasibility.

C. CFO Mixed Doppler Component Extraction

With the spatial and delay-domain parameters now resolved, this subsection details the final stage of our algorithm: extracting the relative Doppler shifts. This is achieved in two steps. First, we reconstruct the signal using the estimated DOA and DPT and extract the CFO-mixed Doppler component for each preamble. Second, by performing a correlation between these extracted components, the common CFO term is canceled out, yielding the desired relative Doppler shifts.

Based on the estimated DOA and DPT, the m -th preamble $\tilde{\mathbf{y}}_{k,m}$ for the k -target could be reconstructed, the formulated signal is defined as $\bar{\mathbf{y}}_{k,m}$, which is shown as

$$\bar{\mathbf{y}}_k = \hat{\mathbf{v}}_k \hat{\mathbf{y}}_k = \alpha_k \tilde{\mathbf{v}}(f_{D,k}) \mathbf{b}(\hat{\theta}_k) \otimes (\mathbf{F}^H(\mathbf{x} \odot \mathbf{r}(\hat{\tau}_k))), \quad (41)$$

where the preamble index m is omitted for the ease of expression, as the estimation is made separately for each preamble m with the corresponding PN one-bit measurements. Define the received signal $\mathbf{y} = \sum_{k=0}^{K-1} \tilde{\mathbf{y}}_k = \sum_{k=0}^{K-1} \hat{\mathbf{v}}_k \hat{\mathbf{y}}_k$, with the elements $y_i = \sum_{k=0}^{K-1} \hat{v}_k \hat{y}_{k,i}$. It is known that the latter part $\hat{\mathbf{y}}_k$ could be reconstructed with the estimated parameters. In this case, the CFO mixed Doppler component $\hat{v}_k = \alpha_k \tilde{\mathbf{v}}(f_{D,k})$ could be regarded as an unknown complex variable, where there are K underestimated variables in total. These parameters could be estimated via the steepest gradient descent method with the convex likelihood function (35) with respect to $\hat{v}_k$. For the k -th target, by taking derivatives of $\hat{v}_k^{\Re}$ and $\hat{v}_k^{\Im}$, the gradients are

$$\Delta \hat{v}_k^{\Re} = \sum_{i=1}^{PN} \left(-f'(\xi_i^{\Re}) \Re\{z_i\} \Re\{\hat{y}_{k,i}\} - f'(\xi_i^{\Im}) \Im\{z_i\} \Im\{\hat{y}_{k,i}\} \right), \quad (42)$$

where $\xi_i^{\Re}$ and $\xi_i^{\Im}$ are denoted as

$$\xi_i^{\Re} = \Re\{z_i\} \left(\hat{v}_k^{\Re} \Re\{\hat{y}_{k,i}\} - \hat{v}_k^{\Im} \Im\{\hat{y}_{k,i}\} + \sum_{j=0, j \neq k}^{K-1} \Re\{\tilde{y}_{j,i}\} \right),$$

$$\xi_i^{\Im} = \Im\{z_i\} \left(\hat{v}_k^{\Re} \Im\{\hat{y}_{k,i}\} + \hat{v}_k^{\Im} \Re\{\hat{y}_{k,i}\} + \sum_{j=0, j \neq k}^{K-1} \Im\{\tilde{y}_{j,i}\} \right), \quad (43)$$

respectively.

And the imaginary part can also be calculated as

$$\Delta \hat{v}_k^{\Im} = \sum_{i=1}^{PN} \left(f'(\xi_i^{\Re}) \Re\{z_i\} \Im\{\hat{y}_{k,i}\} - f'(\xi_i^{\Im}) \Im\{z_i\} \Re\{\hat{y}_{k,i}\} \right). \quad (44)$$

Similarly, the gradients for the Doppler components are derived using the chain rule on the negative log-likelihood function. The rigorous step-by-step derivation is provided in Appendix B. By iteratively updating between K targets via

$$\hat{v}_k^{l+1} = \hat{v}_k^l - \mu \Delta \hat{v}_k. \quad (45)$$

Until convergence, the CFO mixed term is extracted.

After obtaining M groups of extracted data $\hat{\mathbf{v}}(f_{D,k}) = [\hat{v}_{k,0}, \dots, \hat{v}_{k,M-1}]^T \in \mathbb{C}^{M \times 1}$, the CFO-free Doppler frequency vector could be obtained via doing correlation between K targets. The relative Doppler frequency could be obtained by performing the fast Fourier transform with zero padding, from which the power spectrum is calculated as

$$S_{k_i, k_j}(f) = \mathcal{F} \{[(\hat{\mathbf{v}}(f_{D,k_i}) \odot \hat{\mathbf{v}}(f_{D,k_j})^*)^T, \mathbf{0}]\}. \quad (46)$$

The relative Doppler frequency of target k_i and k_j is estimated via

$$\hat{f}_{D,k_i, k_j} = \arg \max_f |S_{k_i, k_j}(f)|. \quad (47)$$

Notice that the absolute Doppler frequency is hard to obtain as the CFO component is a random variable and hard to estimate. However, the relative value is available as the CFO terms for each target are the same. In this case, the absolute value could be calculated with a static reference path, e.g., static clutter, target, anchor, or line-of-sight (LOS) path. Furthermore, we have M groups of DPT, the estimation of delay could be obtained via calculating the mean value, i.e. $\hat{\tau}_k = \frac{1}{M} \sum_{m=0}^{M-1} \hat{\tau}_{k,m}$. Until now, all of the parameters are estimated, and the algorithm is concluded in Algorithm 1.

D. Complexity Analysis

Finally, we evaluate the complexity of the algorithm. To obtain the solution to the unknown parameter θ , one-bit MUSIC is applied, where the main complexity comes from the eigenvalue decomposition of $\mathbf{R}_z$ and derivation of the spectra. Therefore, the computation complexity is $\mathcal{O}(N^3)$. To obtain the solution to the unknown parameter $\tilde{\tau}$, the computation cost for initialization is proportional to $\mathcal{O}(PNKG)$, this computation comes from the matrix multiplication. Also, for each iteration, the computation load from matrix multiplication for gradient calculation is also $\mathcal{O}(PNKG)$, so the overall computation complexity is $\mathcal{O}(R_1 PNKG)$, where R_1 is the number of iterations. There are M preambles so the total complexity is $\mathcal{O}(R_1 PNMGK)$. To obtain the solution to the unknown parameter $\hat{\mathbf{v}}$ for M preambles, the computation complexity is $\mathcal{O}(R_2 R_3 PNMGK)$, where R_2 and R_3 are the iteration number for a specific target and among K targets, respectively. The computation load associated with the final absolute and relative values calculation could be neglected.

Algorithm 1 Parameter Extraction Method for One-Bit Data With Clock Asynchronism

Input: The one-bit measurements $\mathbf{Z}$.

```

1: Generate covariance matrix  $\mathbf{R}_z$  according to (29).
2: SVD:  $\mathbf{R}_z = \mathbf{U}\mathbf{\Sigma}\mathbf{V}$ , where the dimension of  $\mathbf{\Sigma}$  is  $N$ .
3: Generate the spectrum function with null-space vectors.
4: Estimate  $\{\hat{\theta}_k\}_{k=0}^{K-1}$  according to (32).
5: Formulate dictionary matrix  $\mathbf{A}$  according to (36).
6: for each  $m \in [0, M-1]$  do
7:   Initialize  $\mathbf{u}_m = \mathbf{A}^H \mathbf{z}_m$ .
8:   repeat
9:     Compute the gradient according to (40).
10:    Update  $\mathbf{u}_m$ .
11:   until practical convergence
12:   Estimate  $\{\hat{\tau}_{k,m}\}_{k=0}^{K-1}$  according to  $\mathbf{u}_m$ .
13:   Initialize  $\{\hat{\nu}_{k,m} = 0\}_{k=0}^{K-1}$ ,  $\{\hat{\mathbf{y}}_{k,m}\}_{k=0}^{K-1}$  according to (41).
14:   repeat
15:     for each  $k \in [0, K-1]$  do
16:       repeat
17:         Compute the gradient according to (42) and (44).
18:         Update  $\hat{\nu}_{k,m}$ .
19:       until practical convergence
20:     end for
21:   until practical convergence
22: end for
23: Generate  $\hat{\mathbf{v}}(f_{D,k}) = [\hat{\nu}_{k,0}, \dots, \hat{\nu}_{k,M-1}]^T$ .
24: for each  $k \in [0, K-1]$  do
25:   Estimate  $\hat{\tau}_k$  according to  $\hat{\tau}_k = \frac{1}{M} \sum_{m=0}^{M-1} \hat{\tau}_{k,m}$ .
26: end for
27: for  $\frac{K(K-1)}{2}$  target pairs do
28:   Estimate  $\hat{f}_{D,k_{i,j}}$  according to (46) and (47).
29: end for
Output:  $\{\hat{\theta}_k, \hat{\tau}_k\}_{k=0}^{K-1}$ ,  $\{\hat{f}_{D,k_{i,j}}\}_{i=0,j=0}^{K-1}$ .
  
```

Thus the overall complexity of the proposed algorithm is $\mathcal{O}(N^3) + \mathcal{O}(R_1 PNMKG) + \mathcal{O}(R_2 R_3 PNMK)$.

IV. CRAMÉR–RAO LOWER BOUND

This section derives the CRLB for the one-bit OFDM signal with clock asynchronism. While the general Fisher information matrix (FIM) for one-bit quantization has been studied (e.g., [28]), our derivation specifically addresses the singularity issues introduced by the coupled nuisance parameters (δ_τ, δ_f) in the bi-static asynchronous model. To avoid the singularity caused by parameter ambiguity, we derive the bound for the observable composite parameters.

It is known that the CRLB matrix is the inverse of the FIM

$$\mathbf{CRB}(\boldsymbol{\beta}) = \mathbf{J}^{-1}(\boldsymbol{\beta}), \quad (48)$$

where the FIM is obtained via the following equation expressed as

$$\mathbf{J}(\boldsymbol{\beta}) = \mathbb{E} \left[\left(\frac{\partial \ln L(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \right) \left(\frac{\partial \ln L(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \right)^T \right], \quad (49)$$

which is the expectation of the derivative of the log-likelihood function from (22) with respect to unknown variables [28],

[29], [30], [31], [32], [33]. Because all of these samples are independent of each other, the FIM could be rewritten as a summation of the FIM for each sample, i.e.

$$\mathbf{J}(\boldsymbol{\beta}) = \sum_{i=1}^{PNM} \left(\mathbf{J}_i^R(\boldsymbol{\beta}) + \mathbf{J}_i^I(\boldsymbol{\beta}) \right). \quad (50)$$

Take the real part as an example, for one-bit measurements, the expectation is taken over $\Re\{z_i\} = 1$ and $\Re\{z_i\} = -1$. In this case, the expectation equation could be rewritten as

$$\begin{aligned} \mathbf{J}_i^R(\boldsymbol{\beta}) = & \mathbb{P}(\Re\{z_i\} = 1|\boldsymbol{\beta}) \mathbf{J}_i^R(z_i = 1|\boldsymbol{\beta}) \\ & + \mathbb{P}(\Re\{z_i\} = -1|\boldsymbol{\beta}) \mathbf{J}_i^R(z_i = -1|\boldsymbol{\beta}). \end{aligned} \quad (51)$$

Notice $\mathbb{P}(\Re\{z_i\} = 1|\boldsymbol{\beta})$ and $\mathbb{P}(\Re\{z_i\} = -1|\boldsymbol{\beta})$ have been derived from (18) and (19), which are denoted as $\Phi_w(\Re\{\tilde{y}_i\})$ and $\Phi_w(-\Re\{\tilde{y}_i\})$. And in order to obtain FIM $\mathbf{J}_i^R(z_i|\boldsymbol{\beta})$, the derivative of the log-likelihood is calculated first as

$$\frac{\partial \ln L_i^R(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = f'(\Re\{z_i\} \Re\{\tilde{y}_i\}) \Re\{z_i\} \frac{\partial \Re\{\tilde{y}_i\}}{\partial \boldsymbol{\beta}}, \quad (52)$$

where $f'(x) = \phi_w(x)/\Phi_w(x)$ has been defined in (40). Thus the FIM $\mathbf{J}_i^R(z_i = 1|\boldsymbol{\beta})$ is detailed as

$$\begin{aligned} \mathbf{J}_i^R(z_i = 1|\boldsymbol{\beta}) &= \left(\frac{\partial \ln L_i^R(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \right) \left(\frac{\partial \ln L_i^R(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \right)^T \\ &= f'(\Re\{\tilde{y}_i\})^2 \frac{\partial \Re\{\tilde{y}_i\}}{\partial \boldsymbol{\beta}} \frac{\partial \Re\{\tilde{y}_i\}}{\partial \boldsymbol{\beta}^T}. \end{aligned} \quad (53)$$

Similarly, $\mathbf{J}_i^R(z_i = -1|\boldsymbol{\beta})$ is obtained as

$$\mathbf{J}_i^R(z_i = -1|\boldsymbol{\beta}) = f'(-\Re\{\tilde{y}_i\})^2 \frac{\partial \Re\{\tilde{y}_i\}}{\partial \boldsymbol{\beta}} \frac{\partial \Re\{\tilde{y}_i\}}{\partial \boldsymbol{\beta}^T}. \quad (54)$$

Combine the above equations yields

$$\begin{aligned} \mathbf{J}_i^R(\boldsymbol{\beta}) &= \left(\frac{\phi_w^2(\Re\{\tilde{y}_i\})}{\Phi_w(\Re\{\tilde{y}_i\})} + \frac{\phi_w^2(-\Re\{\tilde{y}_i\})}{\Phi_w(-\Re\{\tilde{y}_i\})} \right) \frac{\partial \Re\{\tilde{y}_i\}}{\partial \boldsymbol{\beta}} \frac{\partial \Re\{\tilde{y}_i\}}{\partial \boldsymbol{\beta}^T} \\ &= \frac{\phi_w^2(\Re\{\tilde{y}_i\})}{\Phi_w(\Re\{\tilde{y}_i\})\Phi_w(-\Re\{\tilde{y}_i\})} \frac{\partial \Re\{\tilde{y}_i\}}{\partial \boldsymbol{\beta}} \frac{\partial \Re\{\tilde{y}_i\}}{\partial \boldsymbol{\beta}^T}. \end{aligned} \quad (55)$$

Notice $\frac{\partial \Re\{\tilde{y}_i\}}{\partial \boldsymbol{\beta}}$ is required to calculate. For any index i , the received signal $\tilde{y}_i$ could be further written as

$$\tilde{y}_{p,n,m} = \sum_{k=0}^{K-1} \alpha_k \tilde{v}_m(f_{D,k}) \mathbf{f}_p^H(\mathbf{x} \odot \mathbf{r}(\tilde{\tau}_{k,m})) e^{-j\pi n \sin(\theta_k)}, \quad (56)$$

where $\mathbf{f}_p$ is the p -th column of the IFFT matrix $\mathbf{F}$.

As $\boldsymbol{\beta} = [\boldsymbol{\tau}^T, \boldsymbol{f}_D^T, \boldsymbol{\theta}^T, \boldsymbol{\alpha}_R^T, \boldsymbol{\alpha}_T^T, \boldsymbol{\delta}_\tau^T, \boldsymbol{\delta}_f^T]^T \in \mathbb{R}^{(5K+2M) \times 1}$, the FIM $\mathbf{J} \in \mathbb{R}^{(5K+2M) \times (5K+2M)}$, while it is mentioned in the problem formulation section that TO and CFO introduce ambiguities during parameter estimation, which results in uncertainty in absolute value estimation. From the CRLB aspect, it causes the FIM to become singular and makes it impossible to calculate the inverse matrix $\mathbf{CRB}(\boldsymbol{\beta})$.

In this case, we reconstruct $\boldsymbol{\beta}' = [\tilde{\boldsymbol{\tau}}^T, \boldsymbol{\theta}^T, \hat{\mathbf{v}}_R^T, \hat{\mathbf{v}}_I^T]^T \in \mathbb{R}^{(3M+1)K \times 1}$ as they are the optimized variables in the optimization problem (24). Specifically, $\tilde{\tau}_{k,m} = \tau_k + \delta_{\tau,m} = \tilde{\tau}_{k,m}^c$, range estimation $\tilde{r}_{k,m}$ is considered in CRLB calculation to avoid

TABLE I
SIMULATION SETTINGS

Parameters	Values	Parameters	Values
f_c	4.9 GHz	τ	$\{4.5, 5.5\} \mu\text{s}$
Δf	30 kHz	f_D	$\{49, 171.5\} \text{ Hz}$
P	3333	θ	$\{0.254, 0.487\} \text{ rad}$
T_p	2.5 ms	α	$\{1, 1\}$
M	20		
N	8		
δ_τ	$[-0.1, 0.1] \mu\text{s}$		
δ_f	$[-100, 100] \text{ Hz}$		

matrix singular as the delay estimates are extremely small. The derivatives are computed as follows

$$\frac{\partial \Re\{\tilde{y}_{p,n,m}\}}{\partial \tilde{r}_{k,m}} = \Re\left\{\hat{v}_{k,m} \mathbf{f}_p^H(\boldsymbol{\xi} \odot \mathbf{r}(\tilde{r}_{k,m})) e^{-j\pi n \sin(\theta_k)}\right\}, \quad (57)$$

where $\boldsymbol{\xi} = \mathbf{x} \odot [1, e^{-j2\pi\Delta f/c}, \dots, e^{-j2\pi(P-1)\Delta f/c}]^T$,

$$\frac{\partial \Re\{\tilde{y}_{p,n,m}\}}{\partial \theta_k} = \Re\left\{\hat{v}_{k,m} \mathbf{f}_p^H(\mathbf{x} \odot \mathbf{r}(\tilde{r}_{k,m})) e^{-j\pi n \sin(\theta_k)} \gamma_{k,n}\right\}, \quad (58)$$

where $\gamma_{k,n} = -j\pi n \cos(\theta_k)$,

$$\frac{\partial \Re\{\tilde{y}_{p,n,m}\}}{\partial \hat{v}_{k,m}^R} = \Re\left\{\mathbf{f}_p^H(\mathbf{x} \odot \mathbf{r}(\tilde{r}_{k,m})) e^{-j\pi n \sin(\theta_k)}\right\}, \quad (59)$$

$$\frac{\partial \Re\{\tilde{y}_{p,n,m}\}}{\partial \hat{v}_{k,m}^I} = -\Im\left\{\mathbf{f}_p^H(\mathbf{x} \odot \mathbf{r}(\tilde{r}_{k,m})) e^{-j\pi n \sin(\theta_k)}\right\}. \quad (60)$$

Combining these equations yields

$$\begin{aligned} & \frac{\partial \Re\{\tilde{y}_{p,n,m}\}}{\partial \boldsymbol{\beta}} \\ &= \left[\frac{\partial \Re\{\tilde{y}_{p,n,m}\}}{\partial \tilde{\mathbf{r}}^T} \quad \frac{\partial \Re\{\tilde{y}_{p,n,m}\}}{\partial \boldsymbol{\theta}^T} \quad \frac{\partial \Re\{\tilde{y}_{p,n,m}\}}{\partial \hat{\mathbf{v}}_R^T} \quad \frac{\partial \Re\{\tilde{y}_{p,n,m}\}}{\partial \hat{\mathbf{v}}_I^T} \right]^T. \end{aligned} \quad (61)$$

The FIM is obtained with vector (61), and the CRLB matrix could also be calculated; the imaginary part could be derived similarly and is omitted here.

V. SIMULATION RESULTS

Simulation results are provided in this section to verify the effectiveness and advantages of the method. Performance of the optimization algorithm is verified first and compared with the CRLB baseline. The robustness of the proposed method to clock asynchronism is also tested under different TO and CFO in subsection V-B. The results under different system settings, i.e., antenna and preamble numbers, are presented in subsection V-C. Baseline comparison with other methods is presented in subsection V-D. And subsection V-E illustrates the sensing range improvement under practical settings.

The simulation settings are shown in Table I: the carrier frequency is 4.9 GHz, with 100 MHz bandwidth and the subcarrier interval is 30 kHz, thus the subcarrier number P is equal to 3333; 20 preambles are used for sensing with the PRI of $T_p = 2.5$ ms, and the number of receive antennas is set to 8. Two targets are considered with the parameters $\theta_0 = 0.254$ rad, $\theta_1 = 0.487$ rad, $f_{D,0} = 49$ Hz, $f_{D,1} = 171.5$ Hz, $\tau_0 = 4.5 \mu\text{s}$, $\tau_1 = 5.5 \mu\text{s}$, corresponding to the radial velocity and range components $v_0 = 3$ m/s, $v_1 = 10.5$ m/s,

$r_0 = 136.4$ m, $r_1 = 165.2$ m, respectively. TO and CFO are uniformly distributed from $[-0.1, 0.1] \mu\text{s}$ and $[-100, 100]$ Hz, respectively. In addition, SNR is defined as the transmitting power ratio:

$$\text{SNR} = \sigma_x^2 / \sigma_w^2, \quad (62)$$

which is set to -15 dB without other statements, and the equivalent path loss α is set to $\{1, 1\}$. Root-mean-square error (RMSE) is adopted as a metric to evaluate the performance. Take range estimation as an example. The equation is expressed as

$$\text{RMSE}(r) = \sqrt{\frac{\sum_{l=1}^L \|\mathbf{r} - \hat{\mathbf{r}}\|_2^2}{KL}}, \quad (63)$$

where L is the repetition number of the Monte Carlo experiment.

The simulation parameters are configured to reflect a realistic 5G-A deployment scenario. The carrier frequency, bandwidth, and subcarrier spacing follow the 5G New Radio (NR) specifications for the sub-6 GHz band. The number of antennas is set to $N = 8$ to balance angular resolution with the hardware complexity constraints of low-cost micro-receiving nodes. The clock asynchronism parameters ($T_{\max} = 0.1 \mu\text{s}$, $C_{\max} = 100$ Hz) are selected to model the typical instability of uncalibrated, free-running oscillators in distributed passive nodes [17], [21], creating a challenging scenario where conventional coherent processing fails.

It is worth noting that despite the coarse one-bit quantization and low per-sample SNR, high estimation accuracy is achievable due to the significant processing gain. The system integrates energy across $P = 3333$ subcarriers and $N = 8$ antennas. The total integration gain is approximately $10 \log_{10}(PN) \approx 44.3$ dB. Consequently, an input SNR of -15 dB translates to an effective SNR of nearly 30 dB after processing, which provides a sufficient link budget to support the reported estimation performance.

A. Performance Validation and CRLB Comparison

This subsection demonstrates the performance of the optimization algorithm and compares it with the theoretical CRLB. The estimated parameters are optimization variables in the optimization problem (27), which are not the final sensing parameters: range, radial velocity, and DOA.

Fig. 4 shows the estimation performance of the optimization problem. The root CRLB (RCRLB) results of the optimization variables under one-bit quantization and infinite-bit quantization are derived as benchmarks for each underestimated parameter. To comprehensively evaluate the target resolvability, a multi-target scenario ($K = 2$) is also introduced. The target parameters are set to $\boldsymbol{\theta} = [0.254, 0.504]$ rad, $\mathbf{r} = [13.64, 15.14]$ m, and $\mathbf{v} = [3, 4.5]$ m/s. This setup specifically tests the system at its resolution limits, with the angular separation ($\Delta\theta = 0.25$ rad) aligning with the Rayleigh limit of the 8-element array, and the range separation ($\Delta r = 1.5$ m) matching the theoretical resolution of the 100 MHz bandwidth. Variables r , v and θ in the figure represent the TO mixed range estimates $\tilde{r}_m = \tilde{\tau}_m/c$, Doppler embedded path loss $\hat{v}_m$, and

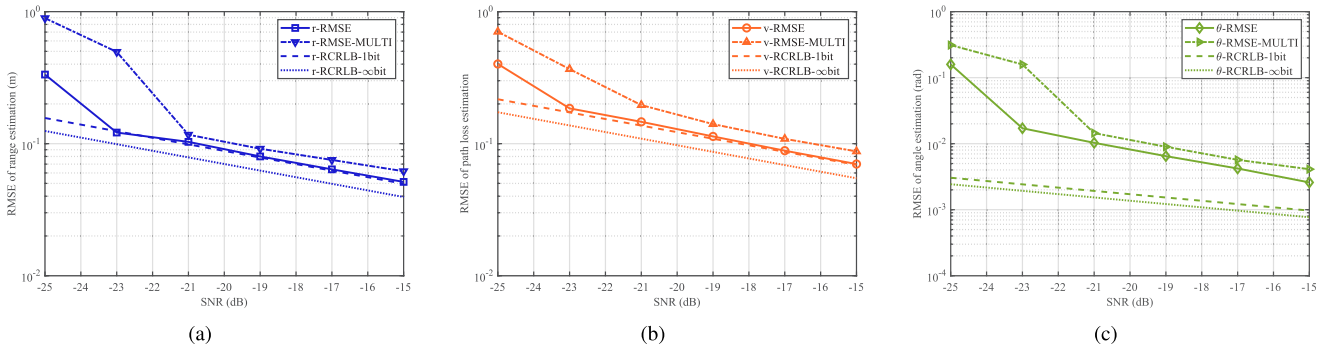


Fig. 4. Estimation performance of the proposed method under different SNRs for (a) to mixed range estimates $\tilde{r}_m = \tilde{r}_m/c$ (b) doppler embedded path loss $\hat{v}_m$ (c) DOA estimates θ .

DOA estimates θ , respectively. The unit of r is m, and that of θ is rad. The simulation results for each preamble m are the same.

The simulation results demonstrate that the proposed method successfully resolves closely spaced targets. As shown in the figures, in the working SNR region, the estimation RMSE for the multi-target case converges to the single-target baseline and tightly tracks the theoretical RCRLB. This indicates that the algorithm effectively utilizes the waveform orthogonality to decouple targets without performance degradation in the operational regime, although a slight performance gap exists at lower SNRs due to the threshold effect.

It is observed that the results of the Monte Carlo experiments approach the RCRLB for $\tilde{r}_m$ and $\hat{v}_m$ estimates, as the maximum likelihood estimation is applied. At the same time, the performance of DOA estimation is worse than the theoretical RCRLB, this is due to the use of one-bit MUSIC for its simplicity. This leads to a seemingly paradoxical observation in Fig. 4: although the DOA estimate is sub-optimal, the subsequent range and path loss estimation stages still achieve near-RCRLB performance. This apparent decoupling occurs because the DOA error's impact manifests as a minor degradation in their effective input SNR. Specifically, the small DOA error results in a minimal array gain loss during coherent accumulation. As quantitatively analyzed in Appendix C, this gain loss is so small (e.g., merely 0.02 dB at an SNR of -21 dB) that the effective SNR remains almost ideal.

Moreover, the CRLB difference between the one-bit and high-precision cases is around 1.96 dB, which corresponds with the theoretical analysis [28]. This small performance gap is acceptable given the great simplification in receiver design, validating that the proposed low-complexity algorithm can approach the theoretical limit with minimal performance loss.

B. Robustness to Clock Asynchronism

This subsection illustrates the impact of clock asynchronism, i.e., varying values of the TO and CFO, on the estimation result. The effects of clock dithering are described first, followed by a simulation over the estimation performance of the absolute values of range, angle estimations, and the relative value of velocity.

Fig. 5(a) illustrates the effects of the TO. Notice that for different preambles, the TO component is time-varying and

randomly dithering around the true value. It is intuitively shown that the TO components for each target are the same. In this case, the absolute value could be obtained with a pre-known fixed reference path. The grid number for range estimation is 400, ranging from 0 m to 400 m. In this case, the sparse vector length is 800 for two-target estimation, i.e. former 400 for target one with estimated DOA $\hat{\theta}_0$ and the latter part for target two with DOA $\hat{\theta}_1$. Thus, energy from antennas could be used in range estimation for coherent accumulation.

Fig. 5(b) and Fig. 5(c) depicts the effects of CFO. These power spectrum figures are generated by doing FFT with $\hat{v}(f_{D,1})$ directly, which is $v_1 = 10.5$ m/s, and after correlation with the other target whose radial velocity is $v_0 = 3$ m/s, respectively. Note that CFO acts like additive noise on the Doppler steering vector, when the maximum value of CFO $C_{\max}$, i.e. $\delta_f \in [-C_{\max}, C_{\max}]$, is small, the absolute value of velocity could be obtained directly, as is shown in Fig. 5(b), while the performance degrades dramatically with the increasing of $C_{\max}$. However, because the CFO is removed via correlating with the other Doppler vector, the difference of radial velocity is obtained, which is $v_1 - v_0 = 7.5$ m/s no matter what the value $C_{\max}$ is, as is shown in Fig. 5(c). Practically, the absolute value of velocity could be obtained via correlating with a static path signal, like ground clutter.

Fig. 6 compares the performance of the proposed method with BS-QOMP, which is a compressive sensing method for one-bit measurements [34], under different clock asynchronism parameters $T_{\max}$ and $C_{\max}$. Here $T_{\max}$ is the maximum value of TO, i.e. $\delta_\tau \in [-T_{\max}, T_{\max}]$. The practical values for $C_{\max}$ and $T_{\max}$ are 100 Hz and 0.1 μ s, respectively. In the simulations, these values measure the severity of the clock asynchronism. The proposed method works well for velocity and DOA estimation, even under extremely severe clock asynchronism, as the algorithm removes the CFO via correlation. However, the performance gets worse for range estimation with larger $T_{\max}$, because the absolute value of the range is obtained by calculating the mean value of the TO mixed range estimation. In comparison, the BS-QOMP method could not handle the one-bit data with clock asynchronism, resulting in poor performance even under small values of $T_{\max}$ and $C_{\max}$. A small performance crossover can be observed for DOA estimation in Fig. 6(c) at very low TO values. This is due to the different architectural trade-offs between the two algorithms.

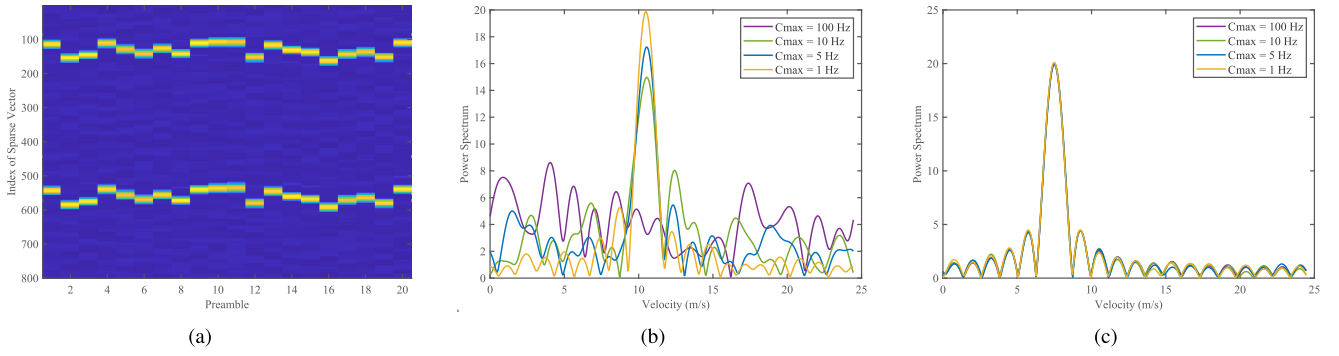


Fig. 5. Figures over (a) effects of to for range estimation illustrated from the varying of sparse vector with 1-bit compressive sensing (b) effects of CFO for Doppler estimation (c) the relative doppler shift component after correlation.

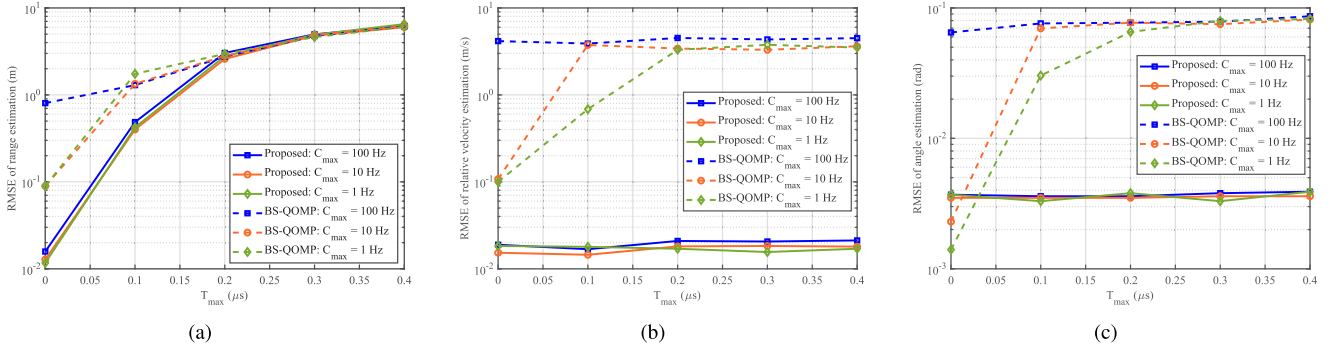


Fig. 6. Estimation performance of the proposed method under different $T_{\max}$ and $C_{\max}$ for (a) absolute value of range (b) relative value of velocity (c) absolute value of DOA.

BS-QOMP, as a joint estimation scheme [34], leverages coherent processing gain across both time and frequency dimensions before estimating the DOA, making it highly effective in near-synchronous conditions. In contrast, our decoupled approach first estimates the DOA using the robust one-bit MUSIC algorithm, which does not utilize this initial time-frequency gain. This design choice prioritizes robustness to the clock asynchronism that is the central challenge of this paper, at the cost of a minor performance loss in the simplified, near-synchronous case. Thus, the traditional method is unable to make accurate estimations under practical conditions, which could be handled well with the proposed algorithm.

C. Analysis of System Parameters

This subsection discovers the effects of system parameters, i.e., antenna number N and preamble number M , on the estimation result.

Fig. 7 demonstrates the algorithm performance under different system settings, i.e., different antenna numbers and preamble numbers. When the array size $N = 4$, the algorithm fails to estimate the target angle, as the DOAs of the two targets are close and they are within the angle resolution of the 4-antenna array. In this case, these two targets could not be distinguished. However, the range estimation remains correct, although without the accumulation gain from the multi-antenna system. Moreover, the velocity embedded vector could not be extracted correctly without the accurate results of the angle, leading to failure in velocity estimation. In addition, the mean value of the DPT is more precise with more preambles, which also improves the accuracy of the velocity result.

D. Comparison With Baseline Methods

This subsection compares the proposed method with other algorithms, i.e., 2D-FFT, CACC [18], CASR [19], and BS-QOMP [34]. The CACC and CASR are two traditional methods designed to cope with clock asynchronism, and the BS-QOMP is a compressive sensing method for one-bit quantization.

Fig. 8 depicts the estimation performance of the proposed method compared against the baselines. It is observed that the 2D-FFT yields the highest error, confirming that standard processing completely fails under the dual impairments of one-bit non-linearity and clock asynchronism. Similarly, the CACC and CASR methods fail to provide accurate estimates. This is because they rely on temporal correlations that are severely deteriorated by the quantization error in the temporal domain, rendering the frequency-domain CSI unreliable for correlation operations. The BS-QOMP method, while capable of handling one-bit measurements, is strongly deteriorated by the clock asynchronism, resulting in a significant error floor.

In comparison, the proposed method successfully estimates the parameters: the absolute DOA and the relative velocity are obtained with high accuracy. It is worth noting that the absolute values of range estimates are influenced by the variance of TO and the number of preambles. Although the optimization result, the variable approaches the RCRLB for the DPT term. However, since the absolute range is obtained by calculating the mean value of the DPT, it is subject to the residual TO variance. As shown by the “Proposed (Perfect Sync)” curve, this performance could be further improved to the theoretical

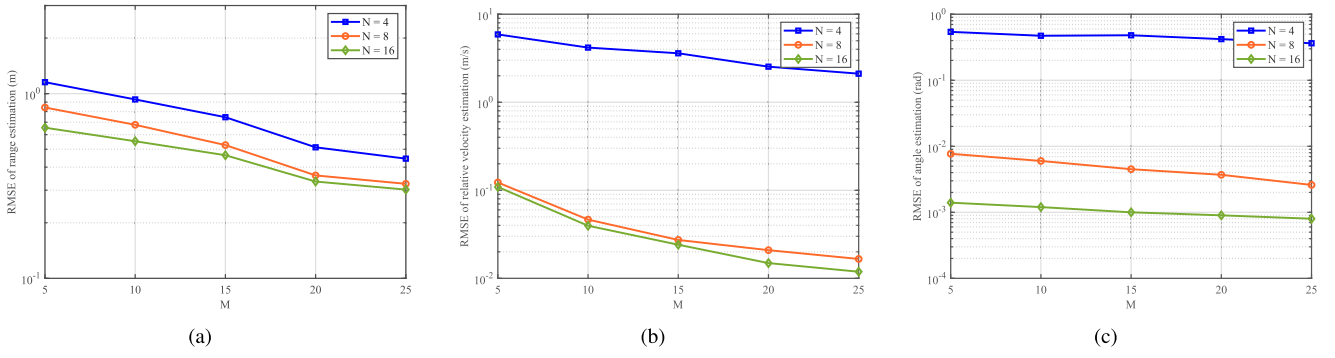


Fig. 7. Estimation performance of the proposed method under different antenna number N and preamble number M for (a) absolute value of range (b) relative value of velocity (c) absolute value of DOA.

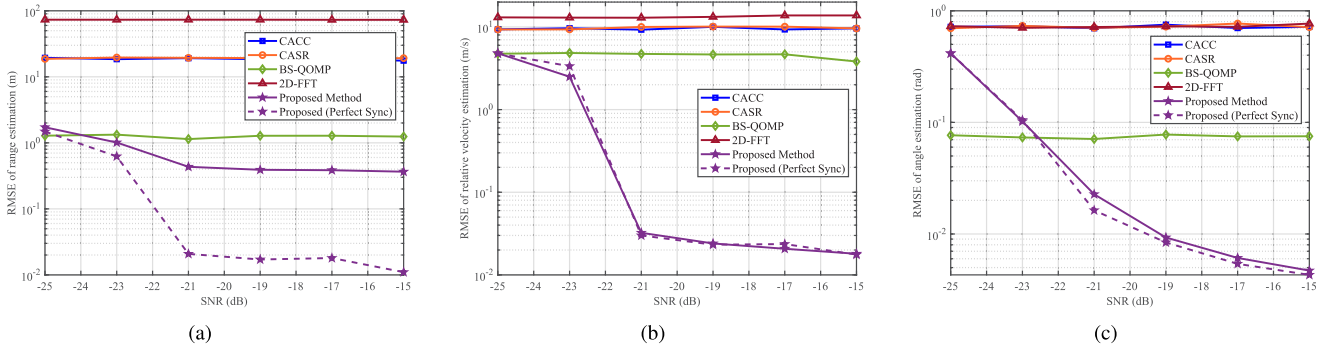


Fig. 8. Estimation performance of the proposed method under different SNRs for (a) absolute value of range (b) relative value of velocity (c) absolute value of DOA.

limit with the help of an additional reference path for TO cancellation.

E. Validation of Range Extension With Practical Settings

This subsection discovers the range extension performance of sensing under practical settings. According to the radar equation

$$P_r = \frac{P_t G_t G_r \lambda^2 \sigma_c}{(4\pi)^3 R_t^2 R_r^2}, \quad (64)$$

where P_t is the transmit power, G_t and G_r are transmit and receive gain, σ_c is the radar cross section of the target, R_t and R_r are distances from the target to the transmit and receive radar, which could be simply calculated via geometry. Besides, the average noise power is expressed as

$$\sigma_w^2 = k_B T_s B_n, \quad (65)$$

where k_B is the Boltzmann constant, T_s is the system noise temperature, and B_n is the noise bandwidth of the receiver. Therefore, the received SNR could be calculated as

$$\text{SNR}_p = P_r / \sigma_w^2, \quad (66)$$

and the specific values of these parameters are detailed in Table II. And other system parameters are as follows in Table I.

Fig. 9 demonstrates the sensing performance without the additional micro receiving node, i.e., a mono-static sensing scenario with the ISAC-BS. The ISAC-BS is located at $\mathbf{p}' = [500, 0]$ m, and the target $\mathbf{p} = [x_p, y_p]$ is randomly located in the two-dimensional space following the uniform distribution, i.e. $x_p \in [0, 1000]$ m, $y_p \in [0, 1000]$ m. The simulation is

TABLE II
PRACTICAL SETTINGS

Parameters	Symbols	Values
Transmit Power	P_t	44 dBm
Transmit Gain	G_t	20 dB
Receive Gain	G_r	18.06 dB
Radar Cross Section	σ_c	0.01 m ²
Boltzmann constant	k_B	1.38×10^{-23} J/K
System Noise Temperature	T_s	290 K
Noise Bandwidth	B_n	100 MHz

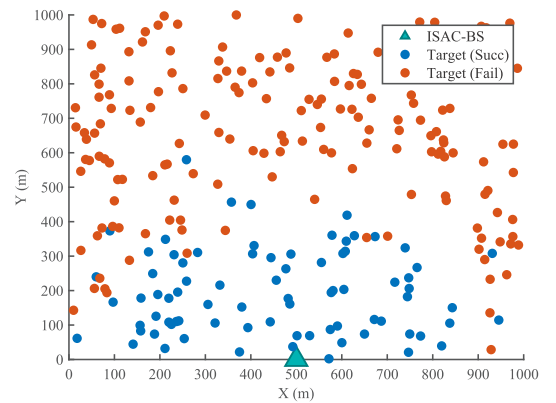


Fig. 9. Mono-static sensing scenario.

repeated for 250 trials, where the blue solid dot represents the range estimation error that is less than 3 m, i.e. target that was successfully estimated, while the orange solid dot stands for the trial that failed. Receive gain is achieved via beamforming, i.e. $G_r = N^2 = 64 = 18.06$ dB, and the naive-FFT algorithm

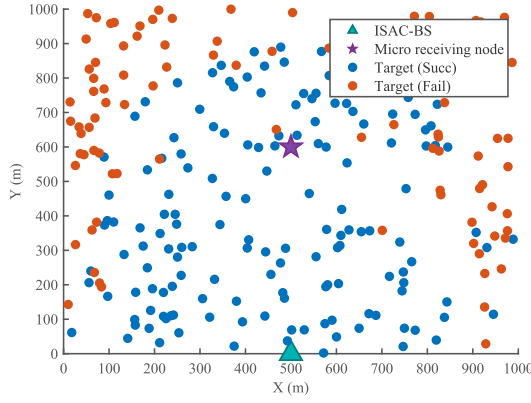


Fig. 10. Bi-static sensing assisted scenario with a micro receiving node.

[35] is applied to estimate the delay component, where the energy is non-coherently integrated from M preambles.

Fig. 10 illustrates the result with an additional micro receiving node located at $\mathbf{p}^r = [500, 600]$ m. The echo signal is processed with the proposed algorithm, where the receive gain G_r is set to 0 dB without analog beamforming. It is shown that the number of successful trials increases, especially for the target located near the micro receiving node, as the receiving SNR increases. The result indicates that, under the practical 5G-A BS setting, the number of successful trials increases from 31.2% to 63.8%. And the sensing range extends from around 400 m to 900 m, increased by 125%, along with the sensing area increases from 280,000 m² to 640,000 m² with the aid of a low-complexity, low-cost one-bit micro receiving node.

VI. CONCLUSION

This paper investigates the one-bit parameter estimation problem with clock asynchronism for low-cost blind spot coverage of the ISAC network. DOAs, delays, and relative Doppler shifts are estimated via one-bit MUSIC, DOA-aided compressive sensing, and CFO mixed Doppler correlation. CRLB has been derived to explain the superiority of the algorithm. Numerical results demonstrate the effectiveness of the proposed method, even in the absence of reference paths. Moreover, simulations with practical settings illustrate the feasibility of the low-cost system for blind spot coverage, which doubles the sensing area and enlarges the sensing range dramatically.

APPENDIX A

DERIVATION OF GRADIENT FOR SPARSE RECOVERY

This appendix details the derivation of the gradient for the sparse recovery problem formulated in Section III-B. The optimization objective is given by:

$$\min_{\mathbf{u}_m} \mathcal{J}(\mathbf{u}_m) = -\ln \mathbb{P}(\mathbf{z}_m | \mathbf{A}\mathbf{u}_m) + \lambda \|\mathbf{u}_m\|_1. \quad (67)$$

A. Gradient of the Log-Likelihood Term

Let $\mathcal{L}_{nll}(\mathbf{u}_m) = -\ln \mathbb{P}(\mathbf{z}_m | \mathbf{A}\mathbf{u}_m)$ denote the negative log-likelihood function. Based on the independent one-bit observations, this term can be expanded as:

$$\mathcal{L}_{nll} = -\sum_{i=1}^{PN} \left(\ln \Phi(u_{m,i}^{\Re}) + \ln \Phi(u_{m,i}^{\Im}) \right), \quad (68)$$

where the arguments of the cumulative distribution function $\Phi(\cdot)$ are defined as:

$$\begin{aligned} u_{m,i}^{\Re} &= \Re\{z_{m,i}\} \Re\{\mathbf{A}\mathbf{u}_m\}_i, \\ u_{m,i}^{\Im} &= \Im\{z_{m,i}\} \Im\{\mathbf{A}\mathbf{u}_m\}_i. \end{aligned} \quad (69)$$

Here, $[\mathbf{v}]_i$ denotes the i -th element of vector $\mathbf{v}$.

We apply Wirtinger calculus to compute the gradient with respect to the conjugate vector $\mathbf{u}_m^*$. Using the chain rule, the gradient is:

$$\frac{\partial \mathcal{L}_{nll}}{\partial \mathbf{u}_m^*} = -\sum_{i=1}^{PN} \left(f'(u_{m,i}^{\Re}) \frac{\partial u_{m,i}^{\Re}}{\partial \mathbf{u}_m^*} + f'(u_{m,i}^{\Im}) \frac{\partial u_{m,i}^{\Im}}{\partial \mathbf{u}_m^*} \right), \quad (70)$$

where $f'(x) = \phi(x)/\Phi(x)$. The partial derivatives of the linear terms with respect to $\mathbf{u}_m^*$ are:

$$\begin{aligned} \frac{\partial \Re\{\mathbf{A}\mathbf{u}_m\}_i}{\partial \mathbf{u}_m^*} &= \frac{1}{2} \mathbf{a}_i^H, \\ \frac{\partial \Im\{\mathbf{A}\mathbf{u}_m\}_i}{\partial \mathbf{u}_m^*} &= \frac{j}{2} \mathbf{a}_i^H, \end{aligned} \quad (71)$$

where $\mathbf{a}_i$ is the i -th row of $\mathbf{A}$. Substituting these back, we obtain the matrix form of the gradient for the data fidelity term:

$$\begin{aligned} \nabla_{\mathbf{u}_m} \mathcal{L}_{nll} &= -\frac{1}{2} \mathbf{A}^H \left(f'(\Re\{\mathbf{z}_m\}) \odot \Re\{\mathbf{A}\mathbf{u}_m\} \odot \Re\{\mathbf{z}_m\} \right. \\ &\quad \left. + j f'(\Im\{\mathbf{z}_m\}) \odot \Im\{\mathbf{A}\mathbf{u}_m\} \odot \Im\{\mathbf{z}_m\} \right). \end{aligned} \quad (72)$$

B. Gradient of the Regularization Term

The second term is the ℓ_1 -norm regularization $\mathcal{R}(\mathbf{u}_m) = \lambda \|\mathbf{u}_m\|_1 = \lambda \sum_i |u_{m,i}|$. Using Wirtinger calculus, the derivative of the modulus $|u|$ with respect to u^* is $\frac{1}{2} \frac{u}{|u|}$. Thus, the gradient is:

$$\frac{\partial \mathcal{R}}{\partial \mathbf{u}_m^*} = \frac{\lambda}{2} \left[\frac{u_{m,1}}{|u_{m,1}|}, \dots, \frac{u_{m,PN}}{|u_{m,PN}|} \right]^T. \quad (73)$$

We define a diagonal matrix $\mathbf{D} = \text{diag} \left\{ \frac{1}{|u_{m,i}|} \right\}_{i=1}^{PN}$. Then, the gradient can be compactly written as:

$$\nabla_{\mathbf{u}_m} \mathcal{R} = \frac{\lambda}{2} \mathbf{D} \mathbf{u}_m. \quad (74)$$

C. Final Gradient Expression

Combining the gradients from the log-likelihood and regularization terms, we arrive at the final steepest gradient descent direction $\Delta \mathbf{u}_m$:

$$\begin{aligned} \Delta \mathbf{u}_m &= -\frac{1}{2} \mathbf{A}^H \left\{ f'(\Re\{\mathbf{z}_m\}) \odot \Re\{\mathbf{A}\mathbf{u}_m\} \odot \Re\{\mathbf{z}_m\} \right. \\ &\quad \left. - \frac{j}{2} \mathbf{A}^H \left\{ f'(\Im\{\mathbf{z}_m\}) \odot \Im\{\mathbf{A}\mathbf{u}_m\} \odot \Im\{\mathbf{z}_m\} \right\} \right. \\ &\quad \left. + \frac{\lambda}{2} \mathbf{D} \mathbf{u}_m \right\}. \end{aligned} \quad (75)$$

This concludes the derivation.

APPENDIX B

DERIVATION OF GRADIENTS FOR DOPPLER EXTRACTION

This appendix details the derivation of the gradients for the CFO-mixed Doppler component extraction in Section III-C. The objective is to minimize the negative log-likelihood function $\mathcal{L} = -\ln \mathbb{P}(\mathbf{z}|\bar{\mathbf{y}})$ with respect to the complex parameter $\hat{v}_k$.

Based on the reconstructed signal model in (41), the total received signal is a linear combination of the target components:

$$\bar{y}_i = \sum_{j=0}^{K-1} \hat{v}_j \hat{y}_{j,i} = \hat{v}_k \hat{y}_{k,i} + \underbrace{\sum_{j \neq k} \hat{v}_j \hat{y}_{j,i}}_{\text{Interference}} \quad (76)$$

where $\hat{y}_{k,i}$ is the known basis derived from the estimated delay and DOA.

The negative log-likelihood function is given by:

$$\mathcal{L} = -\sum_{i=1}^{PN} \left(\ln \Phi(u_i^{\Re}) + \ln \Phi(u_i^{\Im}) \right), \quad (77)$$

where the arguments of the CDF are defined consistent with the one-bit quantization model:

$$\begin{aligned} u_i^{\Re} &= \Re\{z_i\} \Re\{\bar{y}_i\}, \\ u_i^{\Im} &= \Im\{z_i\} \Im\{\bar{y}_i\}. \end{aligned} \quad (78)$$

Let $f'(u) = \phi(u)/\Phi(u)$. By taking the derivative of $\mathcal{L}$ with respect to the real part $\hat{v}_k^{\Re}$, we apply the chain rule:

$$\frac{\partial \mathcal{L}}{\partial \hat{v}_k^{\Re}} = -\sum_{i=1}^{PN} \left(f'(u_i^{\Re}) \frac{\partial u_i^{\Re}}{\partial \hat{v}_k^{\Re}} + f'(u_i^{\Im}) \frac{\partial u_i^{\Im}}{\partial \hat{v}_k^{\Re}} \right). \quad (79)$$

Since $\bar{y}_i$ is linear with respect to $\hat{v}_k$, the partial derivatives of the inner terms are:

$$\begin{aligned} \frac{\partial u_i^{\Re}}{\partial \hat{v}_k^{\Re}} &= \Re\{z_i\} \frac{\partial \Re\{\bar{y}_i\}}{\partial \hat{v}_k^{\Re}} = \Re\{z_i\} \Re\{\hat{y}_{k,i}\}, \\ \frac{\partial u_i^{\Im}}{\partial \hat{v}_k^{\Re}} &= \Im\{z_i\} \frac{\partial \Im\{\bar{y}_i\}}{\partial \hat{v}_k^{\Re}} = \Im\{z_i\} \Im\{\hat{y}_{k,i}\}. \end{aligned} \quad (80)$$

Substituting these into the gradient equation yields the expression for the real part update $\Delta \hat{v}_k^{\Re}$ in (42):

$$\begin{aligned} \Delta \hat{v}_k^{\Re} &= \sum_{i=1}^{PN} \left(-f'(u_i^{\Re}) \Re\{z_i\} \Re\{\hat{y}_{k,i}\} \right. \\ &\quad \left. - f'(u_i^{\Im}) \Im\{z_i\} \Im\{\hat{y}_{k,i}\} \right). \end{aligned} \quad (81)$$

Similarly, for the imaginary part $\hat{v}_k^{\Im}$, we note that:

$$\begin{aligned} \frac{\partial \Re\{\bar{y}_i\}}{\partial \hat{v}_k^{\Im}} &= -\Im\{\hat{y}_{k,i}\}, \\ \frac{\partial \Im\{\bar{y}_i\}}{\partial \hat{v}_k^{\Im}} &= \Re\{\hat{y}_{k,i}\}. \end{aligned} \quad (82)$$

Thus, the gradient with respect to the imaginary part is:

$$\Delta \hat{v}_k^{\Im} = -\sum_{i=1}^{PN} \left(f'(u_i^{\Re}) \Re\{z_i\} (-\Im\{\hat{y}_{k,i}\}) \right.$$

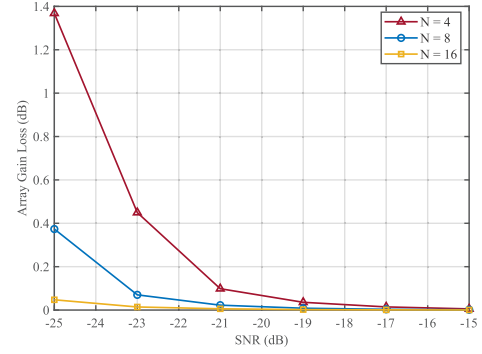


Fig. 11. Effective array gain loss vs. input SNR for different numbers of antennas (N).

$$\begin{aligned} &+ f'(u_i^{\Im}) \Im\{z_i\} \Re\{\hat{y}_{k,i}\}) \\ &= \sum_{i=1}^{PN} \left(f'(u_i^{\Re}) \Re\{z_i\} \Im\{\hat{y}_{k,i}\} - f'(u_i^{\Im}) \Im\{z_i\} \Re\{\hat{y}_{k,i}\} \right). \end{aligned} \quad (83)$$

This matches the gradient expression in (44). The parameter is then updated iteratively via gradient descent.

APPENDIX C

ANALYSIS OF ERROR PROPAGATION FROM DOA ESTIMATION

This appendix provides a quantitative analysis to explain the observation in Fig. 4, where the sub-optimal DOA estimate does not prevent subsequent estimators from achieving near-RCRLB performance. The analysis shows that the error propagation manifests as a minimal degradation in the effective SNR.

The performance penalty arises from the Array Gain Loss (AGL) incurred when beamforming with an imperfect DOA estimate, $\hat{\theta}$. For a ULA with N elements and a true DOA θ , the achieved gain is $G(\hat{\theta}, \theta) = |\mathbf{a}(\hat{\theta})^H \mathbf{a}(\theta)|^2$, where $\mathbf{a}(\cdot)$ is the steering vector. The AGL in decibels is thus defined as:

$$\text{AGL (dB)} = 10 \log_{10}(N^2) - 10 \log_{10}(G(\hat{\theta}, \theta)) \quad (84)$$

For a small estimation error, $\Delta\theta = \hat{\theta} - \theta$, the gain can be approximated by the well-known sinc-like function for a ULA:

$$G(\Delta\theta) \approx \left| \frac{\sin(N\pi \cos(\theta)\Delta\theta/2)}{\sin(\pi \cos(\theta)\Delta\theta/2)} \right|^2 \quad (85)$$

Using Eq. (85), a theoretical calculation for our $N = 8$ array with a representative DOA error of $\Delta\theta = 0.01$ radians (assuming $\theta = 0$) yields a negligible AGL of only 0.012 dB. To validate this under our algorithm, we conducted a simulation measuring the AGL according to Eq. (84), where $\hat{\theta}$ is the actual output of our one-bit MUSIC algorithm at each SNR point. The results are plotted in Fig. 11.

The simulation confirms that the AGL for our chosen $N = 8$ configuration is extremely small, dropping below 0.1 dB for any SNR above -21 dB. For instance, at -21 dB, the simulated AGL is approximately 0.02 dB, in excellent agreement with our theoretical calculation. The figure also reveals that increasing N improves DOA accuracy to such an extent that it consistently outweighs the sensitivity of a narrower beam, resulting in a smaller AGL across all tested SNRs.

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